SafeSense — Architecture Decision Record (draft 2)

Decision: Hazard detection stays server-side in the Flask backend using the three YOLO model files, instead of running inference on-device inside the Flutter app.

Reasoning: On-device inference could not finish inside two seconds on the low-end Android phones used in the pilot, and every model update would have required a Play Store release. Keeping the models behind the Flask API lets us ship a new .pt file without touching the mobile client, and the app only needs the API base URL (configured through dart-define or lib/config/app_config.dart).

Backend owner: Meera Iyer (Flask API and model serving).
Mobile owner: Priya Sharma (Flutter client, release builds).

Approved in the architecture review on 12 March 2026.

SafeSense — Architecture Decision Record (draft 2), continued

Related decision: The Firebase storage bucket used by the first prototype is retired. Uploaded photos are written to local disk on the API host and referenced from the MySQL uploaded_image column, matching the 100% local architecture in the README.

NOTE: This retirement decision was recorded without a named owner. The original author left the team before it was written down, and no review note assigns one. The incoming engineer must confirm who owns revoking the old Firebase credentials.

Open question: video-based hazard detection was proposed during the pilot but was never evaluated or scoped.